

Metadata of the chapter that will be visualized in SpringerLink

Book Title	Uncertainty in Complex Networked Systems	
Series Title		
Chapter Title	Stabilization of Deterministic Control Systems Under Random Sampling: Overview and Recent Developments	
Copyright Year	2018	
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Abstract	This chapter addresses the problem of stabilizing continuous-time deterministic control systems via a sample-and-hold scheme under random sampling. The sampling process is assumed to be a Poisson counter, and the open-loop system is assumed to be stabilizable in an appropriate sense. Starting from as early as mid-1950s, when this problem was studied in the Ph.D. dissertation of R.E. Kalman, we provide a	

historical account of several works that have been published thereafter on this topic. In contrast to the approaches adopted in these works, we use the framework of piecewise deterministic Markov processes to model the closed-loop system, and carry out the stability analysis by computing the extended generator. We demonstrate that for any continuous-time robust feedback stabilizing control law employed in the sample-and-hold scheme, the closed-loop system is asymptotically stable for all large enough intensities of the Poisson process. In the linear case, for increasingly large values of the mean sampling rate, the decay rate of the sampled process increases monotonically and converges to the decay rate of the unsampled system in the limit. In the second part of this article, we fix the sampling rate and address the question of whether there exists a feedback gain which asymptotically stabilizes the system in mean square under the sample-and-hold scheme. For the scalar linear case, the answer is in the affirmative and a constructive formula is provided here. For systems with dimension greater than 1 we provide an answer for a restricted class of linear systems, and we leave the solution corresponding to the general case as an open problem.

Stabilization of Deterministic Control Systems Under Random Sampling: Overview and Recent Developments



Aneel Tanwani, Debasish Chatterjee and Daniel Liberzon

Abstract This chapter addresses the problem of stabilizing continuous-time deterministic control systems via a sample-and-hold scheme under random sampling. The sampling process is assumed to be a Poisson counter, and the open-loop system is assumed to be stabilizable in an appropriate sense. Starting from as early as mid-1950s, when this problem was studied in the Ph.D. dissertation of R.E. Kalman, we provide a historical account of several works that have been published thereafter on this topic. In contrast to the approaches adopted in these works, we use the framework of piecewise deterministic Markov processes to model the closed-loop system, and carry out the stability analysis by computing the extended generator. We demonstrate that for any continuous-time robust feedback stabilizing control law employed in the sample-and-hold scheme, the closed-loop system is asymptotically stable for all large enough intensities of the Poisson process. In the linear case, for increasingly large values of the mean sampling rate, the decay rate of the sampled process increases monotonically and converges to the decay rate of the unsampled system in the limit. In the second part of this article, we fix the sampling rate and address the question of whether there exists a feedback gain which asymptotically stabilizes the system in mean square under the sample-and-hold scheme. For the scalar linear case, the answer is in the affirmative and a constructive formula is provided here. For systems with dimension greater than 1 we provide an answer for a restricted class of linear systems, and we leave the solution corresponding to the general case as an open problem.

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T. Baskar (ed.), *Uncertainty in Complex Networked Systems*,
Systems & Control: Foundations & Applications,
https://doi.org/10.1007/978-3-030-04630-9_6